AI4M TECHNOLOGY

DARK CASCADE FRAMEWORK

Functional Requirements Specification

EnableAI Platform · Autonomous VFFS Intelligence

Document No.	**AI4M-FRS-2604-001**
Title	Dark Cascade Framework — Functional Requirements Specification
Parent URS	AI4M-URS-2604-001 v0.1 (Dark Cascade Framework URS)
Platform	EnableAI · vaenableAI · Dark Cascade Framework
Version	**v0.1 — Section 1 of 15 (Document Control & Metadata)**
Status	**DRAFT — Internal Review | NOT FOR CLIENT RELEASE**
Date	2026-04-04
Author	AI4M Technology — Engineering Team
AI4M Project Lead	Suraj Rathod
Deployment Sites	Site-A: ULP Cavite (Unilever Philippines) · Site-B: HUL Haridwar CLS-1 (HUL India)
Standard	GAMP 5 (2nd Ed.) Appendix D1 · ISO/IEC/IEEE 29148:2018 · IEC 62443 · ISA-88/95 · EARS Syntax
GAMP 5 Category	Category 5 (Custom Application) — Full V-Model validation lifecycle applies

WARNING: *This document is produced section-by-section and reviewed sequentially. Each section is confirmed by the AI4M Engineering Lead before the next section is generated. Do not distribute partial sections externally.*

Section 1 — Document Control & Metadata

1.1 Purpose & Scope of This FRS

This Functional Requirements Specification (FRS) translates the Dark Cascade Framework User Requirements Specification (AI4M-URS-2604-001 v0.1) from user-level 'what' statements into supplier-level 'how' descriptions — defining exactly how each module of the Dark Cascade framework will behave, what workflows it will execute, what data it will produce and consume, and what business rules it will enforce.

This FRS covers the full ten-module Dark Cascade framework applicable across all deployment sites. Site-specific technical parameter bindings (PLC vendor, tag names, camera model, network topology) that vary between ULP Cavite (Site-A) and HUL Haridwar CLS-1 (Site-B) are flagged with [SITE-A] or [SITE-B] labels. Requirements without a site label apply to all deployments.

This FRS is the primary input document for:

- Technical Design Specification (TDS) — component-level engineering
- Factory Acceptance Test (FAT) protocol — OQ-level functional verification
- Site Acceptance Test (SAT) protocol — PQ-level production validation
- GitHub Issues decomposition — Epics, Stories, and Tasks for sprint planning

1.2 Relationship to URS & Traceability Chain

Every FRS requirement in this document traces upward to at least one URS requirement in AI4M-URS-2604-001. The relationship is one-to-many: a single URS requirement typically decomposes into 2–6 FRS requirements that collectively specify how that URS requirement will be satisfied.

TRACEABILITY CHAIN
SIR (Site Investigation Report)
↓ raw observations & client statements
URS AI4M-URS-2604-001 (what the system must do — technology-agnostic)
↓ functional decomposition (this document)
FRS AI4M-FRS-2604-001 (how the system will behave — technology-specific)
↓ architecture & engineering design
TDS AI4M-TDS-2604-001 (component design, DB schema, API contracts)
↓ test protocols
FAT / SAT (functional verification and production qualification)

> **NOTE:** *Every FRS requirement ID carries the format FRS-[MODULE]-[NNN]. The 'Traces To' column in every requirements table maps back to the parent URS ID(s). Orphan FRS requirements (no URS parent) are flagged as gold-plating defects and must be resolved before delivery.*

1.3 Document Conventions

1.3.1 Requirement Qualifier System

Every requirement in this FRS carries exactly one qualifier. The qualifier defines the contractual weight of the requirement and its default priority in sprint planning.

Qualifier	Label	Definition	Consequence of Non-Delivery
MUST	Mandatory — Regulatory / Safety	Non-negotiable. Failure to deliver this requirement means the system cannot be accepted or commissioned. Typically driven by safety, regulatory, or architectural integrity.	Go/No-Go for FAT. System cannot proceed to SAT.
SHALL	Business-Critical	Required for full business value delivery. May be phased to a later sprint if a risk-assessed justification is documented and agreed with the client.	Deferred delivery requires CR and client sign-off. Must be delivered before SAT.
SHOULD	Desirable / Value-Add	Adds measurable value but can be deferred to a later release without preventing commissioning. Backlog item if not included in current sprint plan.	Backlogged. Delivered in next sprint cycle after MUST/SHALL complete.

1.3.2 Requirement Status Codes

Every FRS requirement row carries a Status field. Status governs whether the requirement can be built in the current sprint.

Status	Visual Indicator	Meaning	Action Required
DRAFT	*Green row*	Requirement is fully specified. All inputs confirmed. Ready to be handed to development.	None — include in sprint planning.
TBD-STUB	*Yellow row*	Requirement is written but contains one or more [TBD-xxx] placeholder values. The structure is correct; the value must be confirmed.	Resolve TBD before coding the requirement. Track in Section 15 TBD Register.
BLOCKED	*Red row*	A prerequisite input (client data, hardware confirmation, prior FRS section) has not been received. Requirement cannot be specified yet.	Escalate to Project Lead. Do not start development. Track in Section 15 BLOCKED Register.
APPROVED	*Blue row (bold)*	Requirement reviewed and signed off by both AI4M Tech Lead and client SME. Frozen — changes require a formal Change Request.	Proceed to TDS and FAT protocol authoring.

1.3.3 Confidence & Provenance Tags

Every factual claim in this FRS carries a provenance tag that identifies the source of the stated value. This is the anti-hallucination control: no number, threshold, parameter name, or protocol appears without a tagged source.

Tag	Source Type	Meaning	Example in Document
[CONF]	**Client-Confirmed**	Value explicitly confirmed by client in writing, questionnaire, or meeting minutes.	*Camera trigger delay = 5ms [CONF — ULP questionnaire 2026-03-20]*

Tag	Source Type	Meaning	Example in Document
[INF]	**Inferred**	Value derived logically from confirmed data; inference chain documented.	*Frame rate ≥ 1.83 fps [INF — 110 UPM / 60 = 1.83 sachets/sec]*
[ASS:DOMAIN-NORM]	**Domain Norm Assumption**	Industry-standard practice for this domain. Named basis.	*TLS 1.3 minimum encryption [ASS:DOMAIN-NORM — IEC 62443-3-3 §SR 4.1]*
[ASS:PAST-PROJECT]	**Past Project Basis**	Value reused from a validated prior AI4M project. Project named.	*SQI physics model basis [ASS:PAST-PROJECT — HUL Khamgaon DT, 2025]*
[ASS:EJ]	**Engineering Judgment**	Reasonable technical estimate by AI4M engineer. Must be confirmed at FAT.	*NATS JetStream max_pending = 10,000 messages [ASS:EJ]*
[PENDING]	**Pending Confirmation**	Client response required. Placeholder value used. Tracked in TBD Register.	*max_correction_bound = [PENDING — client to confirm post-commissioning testing]*
[REG]	**Regulatory Baseline**	Mandated by named standard. No client override permitted.	*TLS ≥ 1.3 [REG — IEC 62443-3-3 SR 4.1]*

1.3.4 EARS Syntax Requirement Patterns

All requirements in this FRS use EARS (Easy Approach to Requirements Syntax) patterns. Six patterns are used; each is associated with a trigger keyword.

EARS Pattern	Trigger Keyword	Template & Example
Ubiquitous	**(none)**	The [system] shall [action].Ex: The Edge Gateway shall maintain a tamper-proof audit log.
Event-Driven	**When**	When [trigger event], the [system] shall [action].Ex: When SQI grade is Red for two consecutive cycles, the system shall raise a Major Alarm.
State-Based	**While**	While [system state], the [system] shall [action].Ex: While operating in Offline Mode, the system shall continue SQI computation using the cached Q-BOM.
Conditional	**Where / If**	Where [condition], the [system] shall [action].Ex: Where auto_correction_enabled = FALSE, the system shall display the correction recommendation as an advisory prescription.
Optional Feature	**Where [system] includes**	Where the system includes [feature], the system shall [action].Ex: Where the system includes the Weight Variation module, the system shall acquire piston current data at ≥100Hz.
Unwanted Behaviour	**If / When … shall not**	When [trigger], the [system] shall not [unwanted action].Ex: When a correction exceeds max_correction_bound, the system shall not execute the PLC write.

1.3.5 FRS Requirement ID Format

FRS Requirement IDs follow the pattern: FRS-[MODULE]-[NNN] where MODULE is the 3–5 character module code and NNN is a zero-padded 3-digit sequence number within that module.

Module Code	Section	Module Name	Example FRS IDs
QBOM	§3	Q-BOM Management	*FRS-QBOM-001, FRS-QBOM-002 …*
SQI	§4	Seal Quality Digital Twin	*FRS-SQI-001, FRS-SQI-002 …*
VIS	§5	Visual Inspection (DEOC/CRQS)	*FRS-VIS-001, FRS-VIS-002 …*
CL	§6	Closed-Loop Autonomous Control	*FRS-CL-001, FRS-CL-002 …*
GSM	§7	VFFS Parameter Calculator	*FRS-GSM-001, FRS-GSM-002 …*
ALM	§8	Alert Management	*FRS-ALM-001, FRS-ALM-002 …*
OEE	§9	OEE / BBOEE / Loss Categorization	*FRS-OEE-001, FRS-OEE-002 …*
WGT	§10	Weight Variation Control	*FRS-WGT-001, FRS-WGT-002 …*
SRV	§10	Servo Health Monitoring	*FRS-SRV-001, FRS-SRV-002 …*
EDGE	§11	Edge Infrastructure & Security	*FRS-EDGE-001, FRS-EDGE-002 …*
ICD	§12	Interface Control Document	*FRS-ICD-001, FRS-ICD-002 …*
NFR	§13	Non-Functional Requirements	*FRS-NFR-001, FRS-NFR-002 …*

1.4 Referenced Documents

Ref ID	Document Title	Document Number / Source	Relevance to FRS	Status
RD-001	**Dark Cascade Framework URS**	*AI4M-URS-2604-001 v0.1*	Parent requirements document — all FRS requirements trace to this document	DRAFT
RD-002	**Perfect Sachet URS**	*AI4M-URS-2603-001 v1.0 (PerfectSachet_URS_v1_0.xlsx)*	ULP Cavite site-specific requirements — predecessor document	DRAFT
RD-003	**Perfect Sachet FRS**	*AI4M-FRS-2603-001 v1.1 (PerfectSachet_FRS_v1_1.xlsx)*	Prior FRS for ULP Cavite — pattern reuse for SQI and DEOC modules	DRAFT
RD-004	**Dark Cascade FDD**	*newfdsqbom.docx (Functional Design Document)*	Q-BOM framework concept and operational workflows	Client-Supplied
RD-005	**Dark Cascade Architecture Blueprint**	*architecturalblueprint.docx*	Database schema, workflow architecture, API layer definitions	Client-Supplied
RD-006	**AI4M Project Knowledge Graph**	*AI4M-KG-2603-001 v1.0*	Architecture patterns, estimation benchmarks, issues database ISSUE-001–007	Internal
RD-007	**ULP FRS/TDS Questionnaire**	*ULP_FRS_TDS_Questionnaire_.md*	Client question-answer record — source for [CONF] tags	Internal
RD-008	**GAMP 5 2nd Edition**	*ISPE, 2022*	Validation lifecycle, software categorisation, FRS structure guidance	External Standard
RD-009	**ISO/IEC/IEEE 29148:2018**	*IEEE / ISO, 2018*	Requirements engineering, URS/FRS structure, EARS syntax basis	External Standard
RD-010	**IEC 62443-3-3:2013**	*IEC, 2013*	OT/IT security zone requirements, PLC write access control	External Standard
RD-011	**ISA-88 / ISA-95**	*ISA, 2010/2013*	Equipment hierarchy, Purdue Model mapping, production data model	External Standard
RD-012	**Dark Cascade Gap Analysis**	*Gap register AI4M-FRS-2604-GAP-v1.0*	20-item gap register from URS→FDD cross-reference — all gaps addressed in this FRS	Internal

1.5 Human Review Gates & Sign-Off Record

The Dark Cascade FRS is reviewed and approved section-by-section through a structured gate process aligned to the GAMP 5 V-Model. Each gate must be formally passed before the next section is distributed or the development team is permitted to act on that section's requirements.

Gate	Trigger / When	AI4M Reviewer	Client Reviewer	Due By	Status
G-00	Section 1 (Document Control) — FRS conventions agreed	Suraj Rathod	—	2026-04-07	**PENDING**
G-01	Section 2 (Architecture) — C4 diagrams and tech stack confirmed	AI4M Architect	—	2026-04-10	**PENDING**
G-02	Section 3 (Q-BOM) — Data model and API contract reviewed	AI4M ML Lead	ULP/HUL IT Lead	2026-04-14	**PENDING**
G-03	Section 4 (SQI Engine) — Physics model spec reviewed	AI4M ML Lead	ULP/HUL QA Manager	2026-04-17	**PENDING**
G-04	Section 5 (Vision) — Camera pipeline and defect taxonomy reviewed	AI4M CV Lead	ULP/HUL QA Manager	2026-04-21	**PENDING**
G-05	Section 6 (Closed-Loop) — PLC write workflow and bounds reviewed	AI4M Architect + Client OEM	HUL/ULP IT/OT Manager	2026-04-24	**PENDING**
G-06	Section 7 (GSM App) — Changeover workflow UX reviewed	AI4M ML Lead	ULP Supervisor	2026-04-28	**PENDING**
G-07	Section 8–9 (Alerts + OEE) — Dashboard and loss model reviewed	AI4M Tech Lead	Operations Manager	2026-05-02	**PENDING**
G-08	Section 10–11 (Weight + Edge) — Sensor and security spec reviewed	AI4M Architect	Site IT Manager	2026-05-06	**PENDING**
G-09	Sections 12–13 (ICD + NFR) — All interfaces and perf targets reviewed	AI4M Architect	Site IT Manager	2026-05-09	**PENDING**
G-10	Section 14–15 (Traceability + TBD Register) — Full FRS integrity check	Suraj Rathod	Client SME Lead	2026-05-12	**PENDING**
G-FINAL	Full FRS sign-off — frozen for TDS authoring	Suraj Rathod	Client Sponsor	2026-05-15	**PENDING**

> **WARNING:** *Review gates are sequential. Sections 3–6 (functional core) require client SME co-review for requirements affecting PLC integration and QA thresholds. All gate completions must be documented with: reviewer name, date, comments, and PASS / CONDITIONAL / FAIL outcome before the next section enters draft.*

1.6 FRS Section Index — Full Document Plan

This table serves as the master delivery plan for the complete FRS. Each section is generated and reviewed independently. Column 'FRS Req Count (est.)' indicates the planned number of FRS requirement rows in that section.

§	Section Title	Key Contents	FRS Reqs (est.)	Status	Gate	Blocks TDS?
1	**Document Control & Metadata**	Conventions, referenced docs, review gates, BLOCKED/TBD register stub	—	DRAFT	G-00	NO
2	**System Architecture & C4 Diagrams**	C4 Context, Container, Component; Purdue Model mapping; tech stack; data flow sequence diagrams	—	PENDING	G-01	YES
3	**Q-BOM Functional Specification**	Data model (4 entities), state machine (5 states), REST API contract (7 endpoints), 3 workflows	18	PENDING	G-02	YES
4	**SQI Engine Functional Specification**	Data acquisition pipeline, physics model spec, grading engine, failure mode classifier, report generator	16	PENDING	G-03	YES
5	**Visual Inspection (DEOC/CRQS)**	Camera pipeline, trigger sync, CRQS taxonomy (8 defect classes), DEOC grader, historian, BOM check	18	PENDING	G-04	YES
6	**Closed-Loop Autonomous Control**	Bounds engine workflow, PLC write protocol, audit log spec, write-verification, advisory mode	14	PENDING	G-05	YES
7	**VFFS Parameter Calculator (GSM App)**	QR scan workflow, AI compute service spec, Draft Q-BOM creation, changeover push, UI page spec	12	PENDING	G-06	YES
8	**Role-Based Alert Management**	Dual stream routing, WebSocket delivery spec, alarm escalation state machine, Clear-OK workflow, historian	14	PENDING	G-07	YES
9	**OEE / BBOEE / Loss Categorization**	OEE formula engine, BBOEE rolling calc, stoppage classifier, empty-mat waste formula, shift report	12	PENDING	G-07	YES
10	**Weight Variation & Servo Health**	Piston current acquisition, density ML model, WVI KPI, servo correlation model, alerts	8	PENDING	G-08	YES
11	**Edge Infrastructure & Security**	Edge Gateway spec, Redis offline cache, IP-gating, IEC 62443 zone map, RBAC matrix, TLS spec	12	PENDING	G-08	YES
12	**Interface Control Document (full)**	10 interfaces — protocol, schema, message format, latency, retry, error handling	30	PENDING	G-09	YES
13	**Non-Functional Requirements (detailed)**	Latency budgets per function, availability SLAs, security matrix, environmental spec	14	PENDING	G-09	YES

§	Section Title	Key Contents	FRS Reqs (est.)	Status	Gate	Blocks TDS?
14	**Traceability Matrix (full bidirectional)**	URS→FRS→TDS→Test Case chain for all 47+ requirements	—	PENDING	G-10	YES
15	**BLOCKED & TBD Register**	All open items, dependency owner, resolution target, unblocking condition	—	PENDING	G-10	NO

> **NOTE:** *Estimated FRS requirement count across all sections: 148–170 individual requirement rows, plus workflow tables, state machine tables, data model tables, and API contract tables. Total document when complete: approximately 80–100 pages (A4 landscape).*

1.7 Issue Prevention Checklist (AI4M Standard — ISSUE-001 through ISSUE-007)

The following checklist must be verified against each section of this FRS before it is released for development. Items are drawn from the AI4M Project Knowledge Graph issues database (ISSUE-001 through ISSUE-007). A checked item means the relevant section of the FRS explicitly addresses the mitigation.

Issue ID	FRS Section (s)	Risk Description	FRS Mitigation	Status	Open Action
ISSUE-001	§6, §12	PLC integration scope not formally defined — AI4M writes to unapproved registers	FRS-CL-002 defines approved register whitelist. PLC Data Readiness Checklist required before §6 DRAFT status. DEP-002 in URS §11.2.	*PENDING — DEP-002 not received*	Suraj Rathod to chase site IT/OT by 2026-04-09
ISSUE-002	§5, §7	Camera mount feasibility not confirmed — space constraints may prevent planned inspection points	FRS-VIS-004 includes Camera Feasibility Gate as a prerequisite to §5 APPROVED status. Diagonal mount option specified as fallback in TBD-003.	*PENDING — ASS-006*	AI4M CV Lead to complete site survey by 2026-04-14
ISSUE-003	§11, §13	Production environment spec — IP rating, temperature, vibration not in FRS NFR	FRS-NFR (§13) will include URS-SEN-003 environmental NFR: 15–45°C, RH 20–90%, IP54 minimum.	*Addressed in §13 (PENDING)*	—
ISSUE-004	§4	SQI thresholds deferred to commissioning — FRS grading logic uses placeholder values	FRS-SQI-004 uses configurable Q-BOM thresholds. TBD-STUB status retained until QA sign-off per DEP-004.	*TBD-STUB — DEP-004*	AI4M ML Lead + Site QA to complete calibration protocol
ISSUE-005	§12	Data communication protocol not confirmed for all PLC vendors (especially HUL Mitsubishi)	FRS-ICD-001 specifies multi-protocol adapter: pylogix (AB), snap7 (Siemens), MC Protocol (Mitsubishi), pyomron (Omron). Site-specific binding confirmed in §12 per site.	*Partially addressed — HUL Mitsubishi MC Protocol TBD*	HUL SIR to confirm Mitsubishi tag list
ISSUE-006	§6, §11	Client IT/OT dependencies not mapped with escalation paths	DEP-001 through DEP-007 in URS §11.2. Each has an escalation path to Suraj Rathod. §6 BLOCKED until DEP-002 received.	*Tracked — 3 of 7 open*	Suraj Rathod owns escalation
ISSUE-007	§4	Domain-specific model data requirements not defined upfront	FRS-SQI-001 defines data acquisition at per-cycle level. FRS-SQI physics model requires laminate thermal property library per SKU — sourced from	*Partially addressed*	FRS §3 Q-BOM to formally define laminate data onboarding workflow

Issue ID	FRS Section (s)	Risk Description	FRS Mitigation	Status	Open Action
			Q-BOM. Data collection protocol for new SKUs in FRS-QBOM-007.		

1.8 BLOCKED & TBD Register — Master Stub

This register is the master tracking table for all open items in the FRS. It is populated as each section is drafted and will be fully populated by Section 15. Items marked BLOCKS-FRS-SECTION = YES prevent the corresponding section from reaching DRAFT status and must be resolved before the section gate can pass.

> **NOTE:** *The entries below are pre-populated from the URS TBD Open Register and Gap Analysis. Additional TBD and BLOCKED items will be added as each FRS section is drafted. This register is the single source of truth for open items — every [PENDING] tag in the FRS body references an entry in this table.*

TBD /BLK Ref	FRS Section(s)	Type	Description	Input Required From	Blocks FRS Section?	Priority	Target Resolution
TBD-001	§3 QBOM	**TBD**	max_correction_bound initial default value per SKU family (seal temp, stroke, puller speed separately)	Client / AI4M ML Lead post-commissioning	*§6 CL*	**CRITICAL**	Phase 1 commissioning + 30 days
TBD-002	§4 SQI	**TBD**	SQI Green/Amber/Red threshold values — formal calibration with ULP/HUL QA sign-off	Site QA Manager + AI4M ML Lead	*§4 SQI-004*	**CRITICAL**	Phase 1 commissioning sign-off
TBD-003	§5 VIS	**BLOCKED**	Camera model, resolution, frame rate, FOV, standoff distance per inspection point — ULP Cavite only (FLIR thermal: GAP-003)	AI4M CV Lead + Client Maintenance	*§5 VIS camera spec*	**HIGH**	Camera feasibility gate 2026-04-14
TBD-004	§5 VIS	**TBD**	False Positive / False Negative threshold targets for DEOC grades — client QA to define acceptable rates	Site QA Manager	*§5 FRS-VIS-002*	**HIGH**	Before FAT protocol authoring
TBD-005	§6 CL	**BLOCKED**	PLC Data Readiness Checklist — approved register list for autonomous writes, signed by client IT/OT + PLC OEM	Client IT/OT Manager + PLC OEM	*§6 FRS-CL-002*	**CRITICAL**	DEP-002 — target 2026-04-09
TBD-006	§9 OEE	**BLOCKED**	PLC status code register per site — Strategic vs Non-Strategic categorization mapping	Site IT/OT Manager	*§9 FRS-OEE-003*	**HIGH**	DEP-001 — target 2026-04-09
TBD-007	§10 WGT	**TBD**	Piston current sensor hardware confirmation — installed or procurement required per site	Client Maintenance / Electrical Lead	*§10 FRS-WGT-001*	**HIGH**	DEP-005 — target 2026-04-21
TBD-008	§11 EDGE	**TBD**	Edge offline resilience duration — minimum hours of autonomous operation without server connectivity	Client Operations Manager	*§11 FRS-EDGE-003*	**HIGH**	Confirm with Ops by 2026-04-14

TBD /BLK Ref	FRS Section(s)	Type	Description	Input Required From	Blocks FRS Section?	Priority	Target Resolution
TBD-009	§11 EDGE	**TBD**	HUL Haridwar regulatory scope — whether GMP / 21 CFR Part 11 audit trail validation applies	HUL QA / Regulatory Affairs	*§11 FRS-EDGE (audit log spec)*	**HIGH**	DEP-007 — target 2026-04-21
TBD-010	§5 VIS	**TBD**	HUL Haridwar machine speed (UPM) per VFFS loop — determines camera frame rate spec	HUL Site Engineering	*§5 FRS-VIS-001*	**MEDIUM**	HUL SIR response
TBD-011	§12 ICD	**TBD**	HUL Haridwar Mitsubishi PLC tag list (Loop 4) — required for MC Protocol ICD specification	HUL Site IT/OT + AI4M integration	*§12 FRS-ICD-001 (HUL site binding)*	**HIGH**	HUL SIR response
TBD-012	§5 VIS	**TBD**	Colour deviation tolerance (ΔE threshold) for laminate colour inspection — client quality standard	Site QA Manager	*§5 FRS-VIS-010*	**MEDIUM**	Before FRS §5 APPROVED

1.9 Section 1 Approval & Sign-Off

By signing below, the reviewer confirms that: (a) document conventions are understood and agreed; (b) the section index and delivery plan are accepted; (c) the BLOCKED/TBD register entries are acknowledged; and (d) Gate G-00 is approved, permitting generation of Section 2 (System Architecture).

Role	Name	Signature	Date / Status
AI4M Project Lead	Suraj Rathod	_____	PENDING
AI4M Engineering Lead	[TBD]	_____	PENDING
Client Reviewer (ULP)	Nikki Mamplata	_____	FOR INFORMATION — not required at Section 1 gate
Client Reviewer (HUL)	Nitya Aggrawal / Aarushi Rustagi	_____	FOR INFORMATION — not required at Section 1 gate

1.10 Version History

Version	Date	Author	Status	Description
v0.1	2026-04-04	AI4M Technology	**DRAFT**	Initial issue. Section 1 of 15. Document Control & Metadata, conventions, referenced documents, human review gates, issue prevention checklist, and BLOCKED/TBD register stub populated from URS v0.1 gap analysis (12 open items: 3 BLOCKED, 9 TBD).

> **NOTE:** *Subsequent version entries will be added as sections are confirmed and merged into the master FRS document. Each confirmed section increments the minor version (v0.2, v0.3…). Full document v1.0 will be issued when all 15 sections pass their gates and the final G-FINAL sign-off is recorded.*